NeuralPOM

Supplementary Figures

New Wine in an Old Bottle — NeuralPOM: A Hybrid Ocean Forecast Model with Differentiable Dynamics and Staged Learning.

18 October 2026

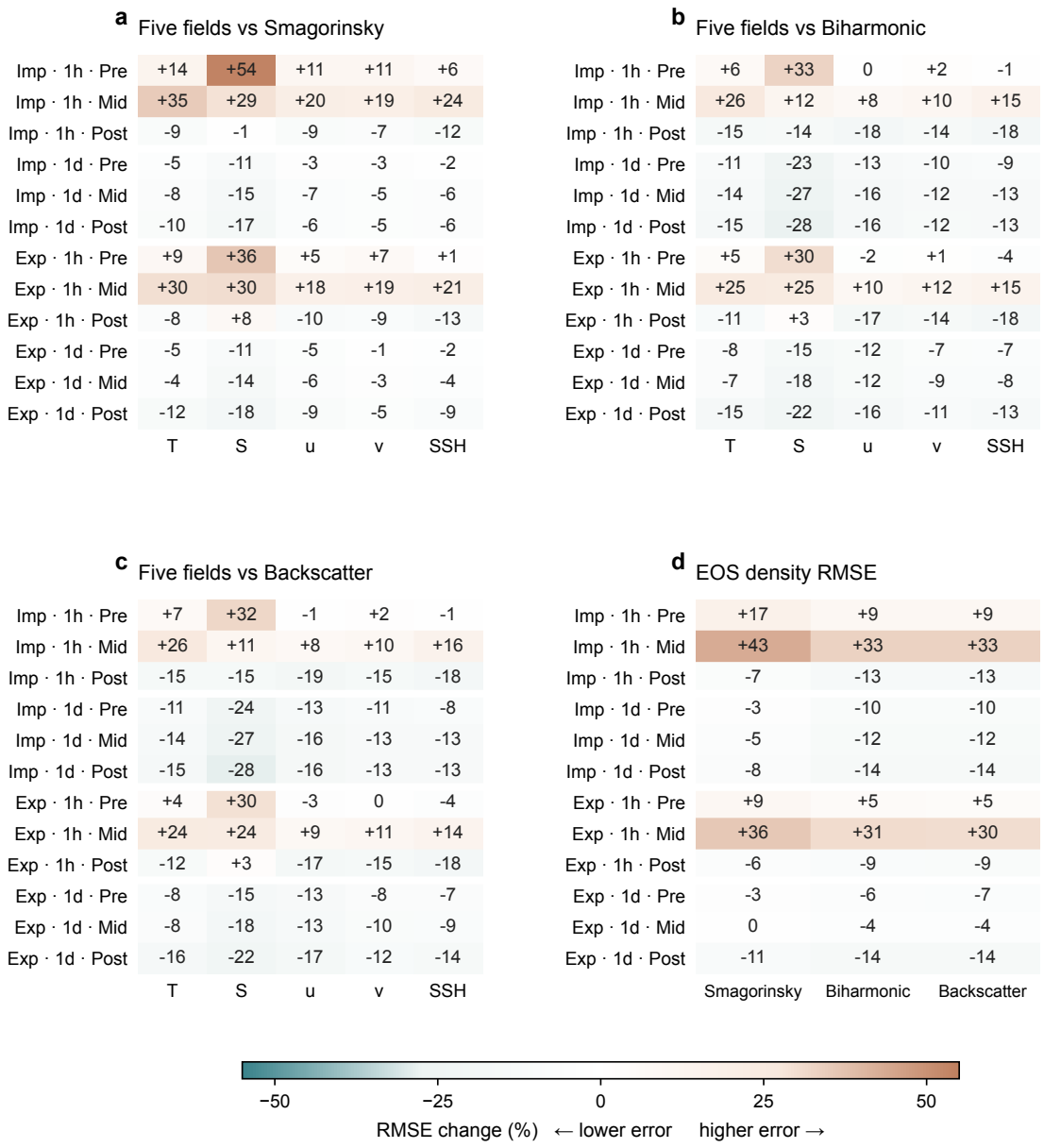


Figure S1: Complete physical-control comparisons over days 1–100. The twelve frozen hybrid configurations under DND are compared with `Smagorinsky`, `Biharmonic` and `Backscatter` for all five forecast fields and EOS-derived density. Numbers are percentage changes in pooled RMSE across the same 16 origins; negative values denote lower error. This is the full matrix underlying the compact production-core comparison in the main text.

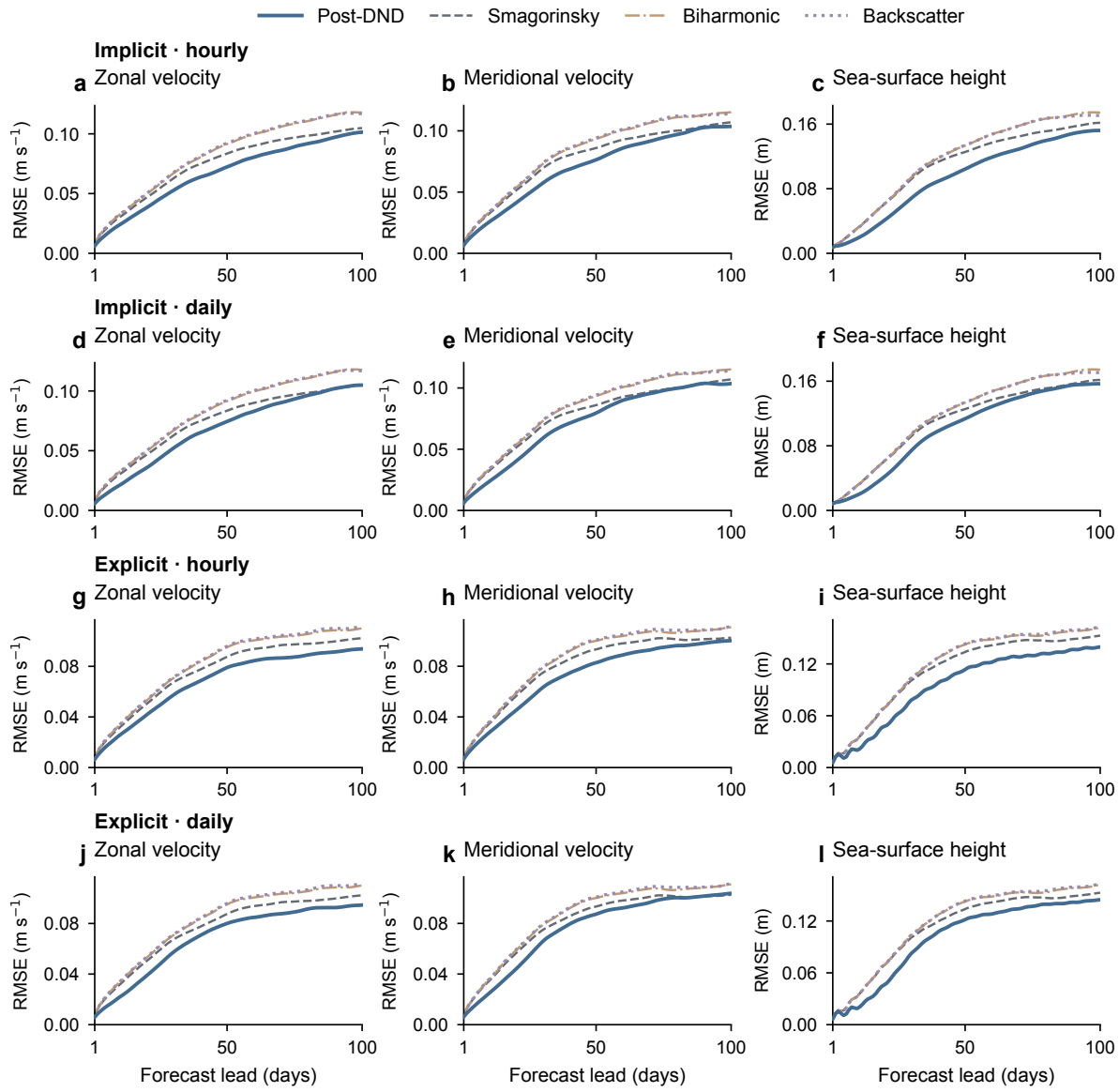


Figure S2: Posttrain-DND horizontal velocity and sea-surface height error growth. Daily RMSE is the square root of mean MSE across 16 overlapping origins at each lead, over days 1–100. Curves show the two cores and correction intervals with matched physical parameterizations (`Smagorinsky`, `Biharmonic` and `Backscatter`). They are unsmoothed; the origins are paired comparisons from one training seed.

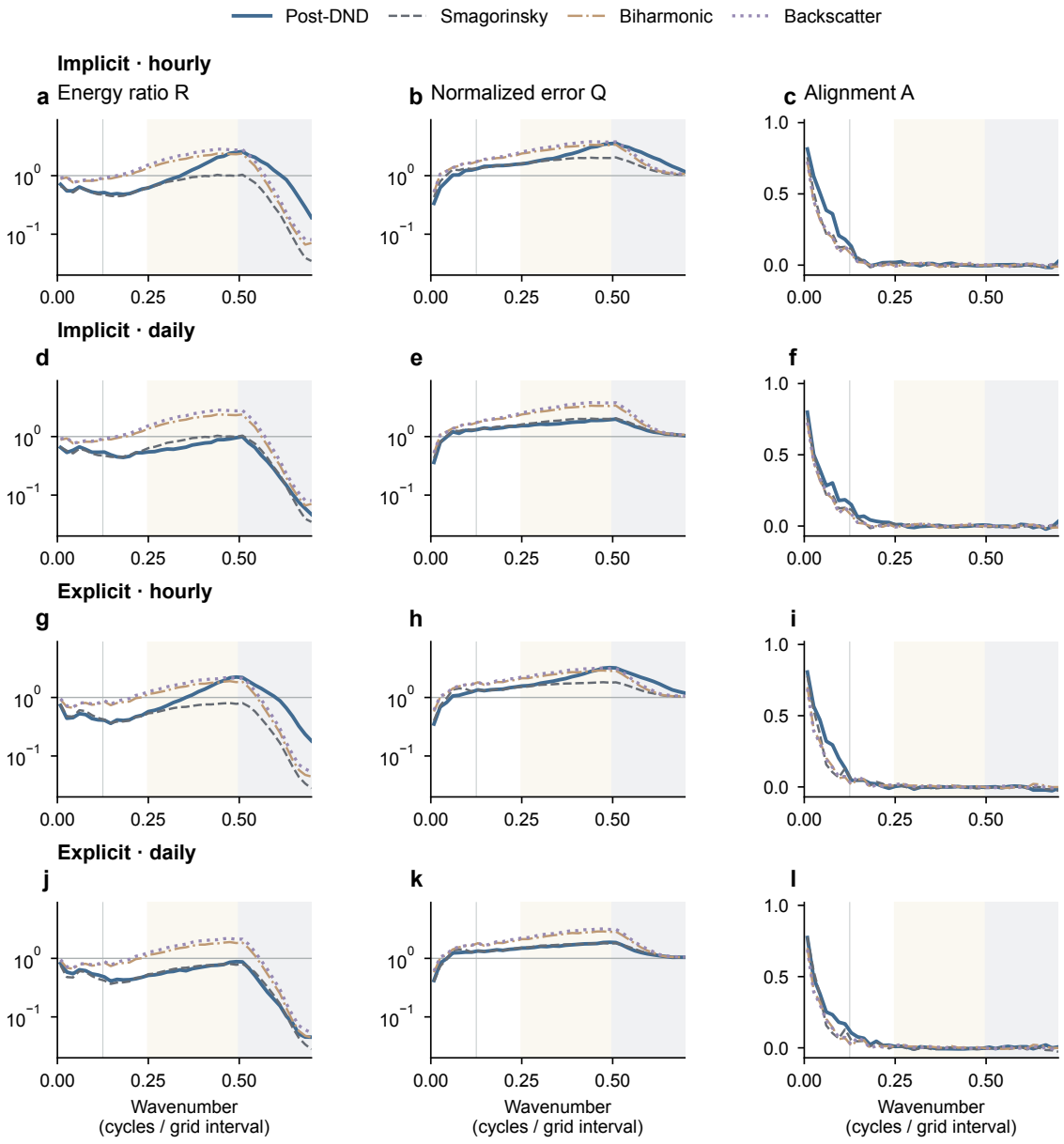


Figure S3: Posttrain-DND spectra, days 51–100. Prediction/reference energy ratio R, error/reference energy ratio Q, and normalized real cross-energy alignment A. The representation is sigma-weighted depth-mean velocity. Energies are pooled over days and origins before ratios are formed. Wavenumber is cycles per grid interval; guides mark 0.125, 0.25 and 0.5. Near-grid and radial corner bins remain visible. Layerwise and depth-mean spectra measure different aspects of the circulation.

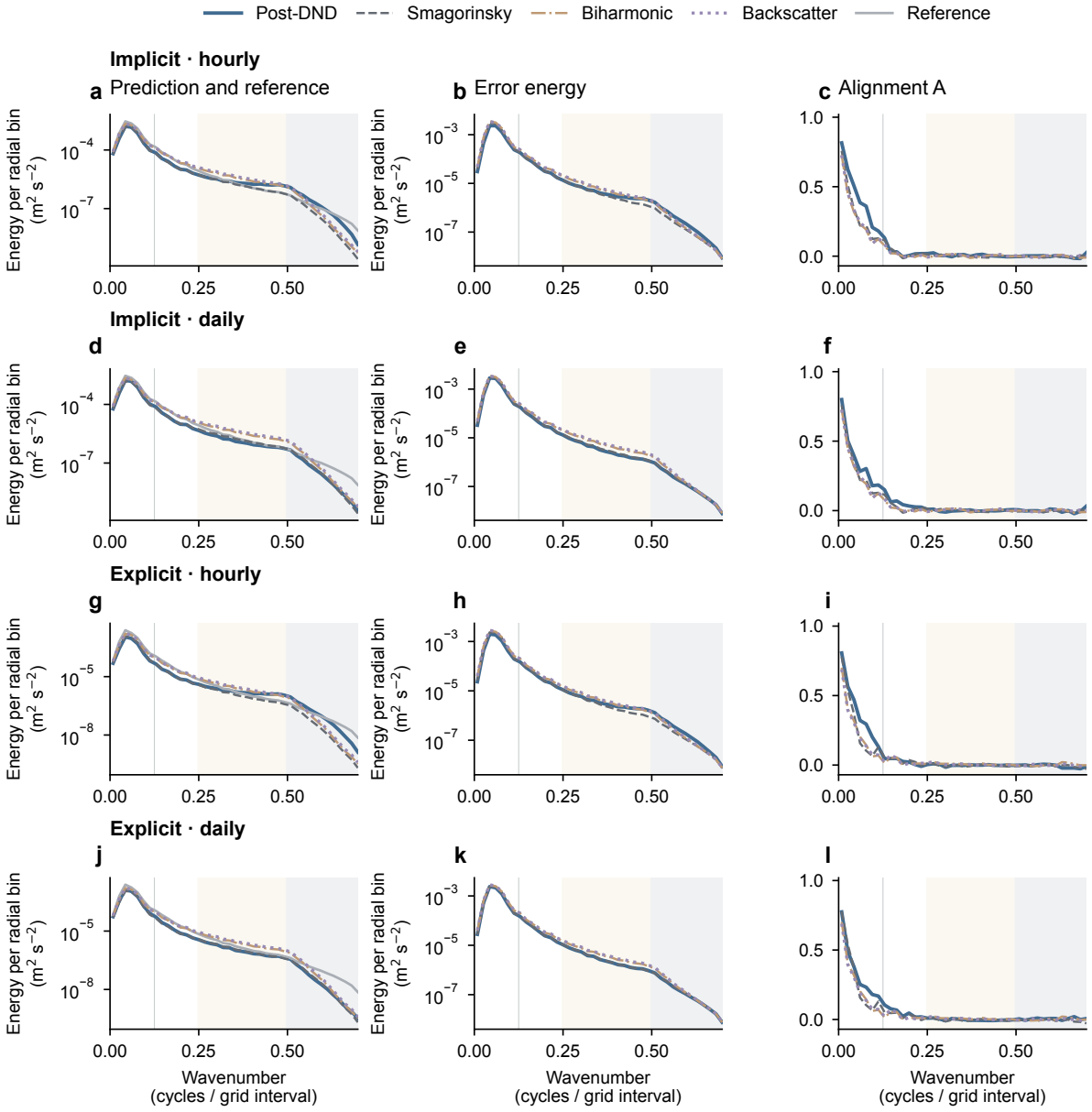


Figure S4: Posttrain-DND spectra, days 51–100. Absolute binned prediction/reference and error energies, with normalized cross-energy alignment. Each radial bin contains total energy in $\mathrm{m}^2\,\mathrm{s}^{-2}$, rather than a continuous spectral density. The representation is sigma-weighted depth-mean velocity. Energies are pooled over days and origins before ratios are formed. Wavenumber is cycles per grid interval; guides mark 0.125, 0.25 and 0.5. Near-grid and radial corner bins remain visible. Layerwise and depth-mean spectra measure different aspects of the circulation.

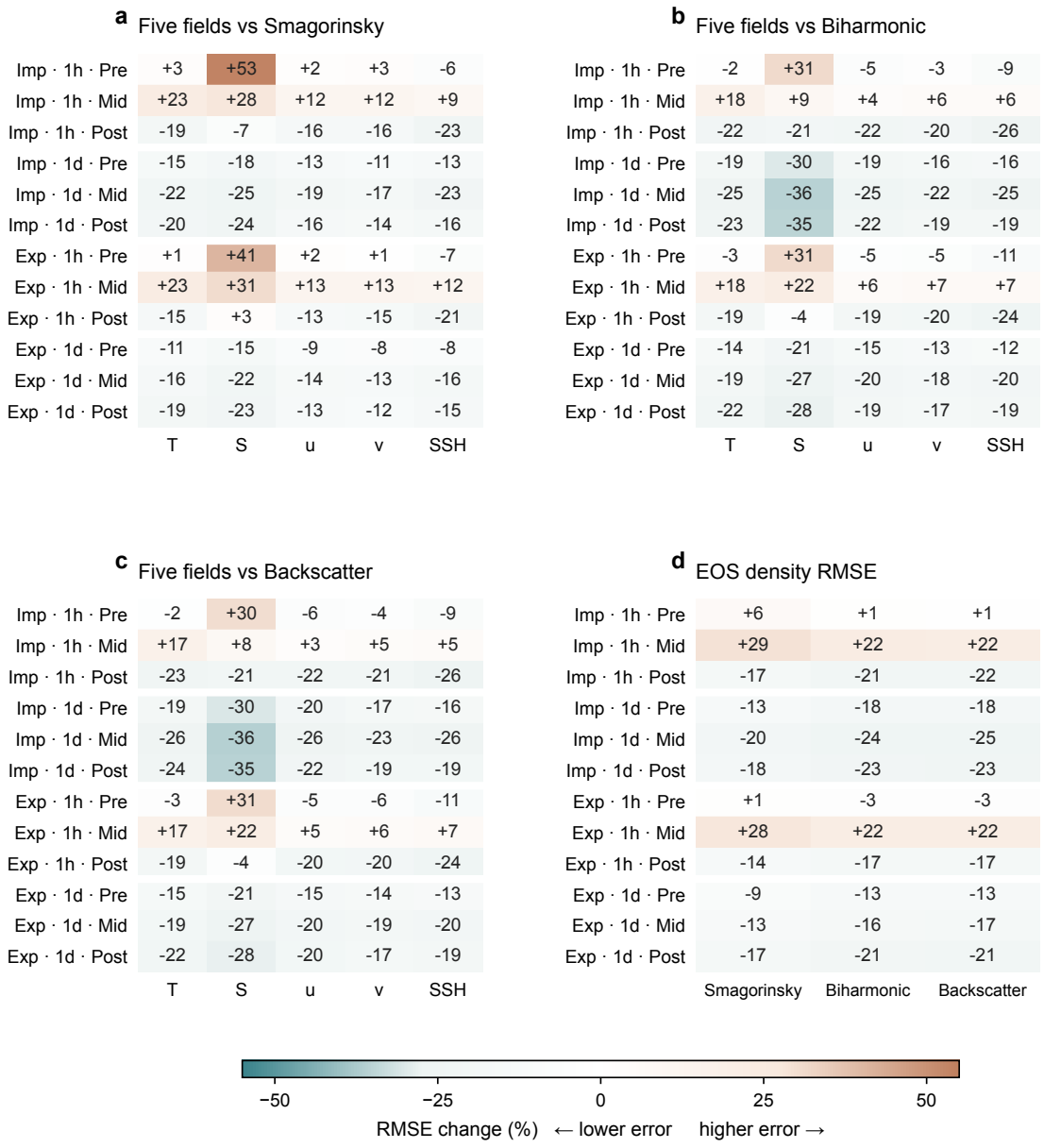


Figure S5: Forecast skill across all frozen stages, days 1–50. Percentage change in RMSE relative to each subgrid parameterization (`Smagorinsky`, `Biharmonic` and `Backscatter`) for all 12 DND configurations. Each score pools daily MSE over the 16 origins and the indicated window before taking its square root. Negative values denote lower error.

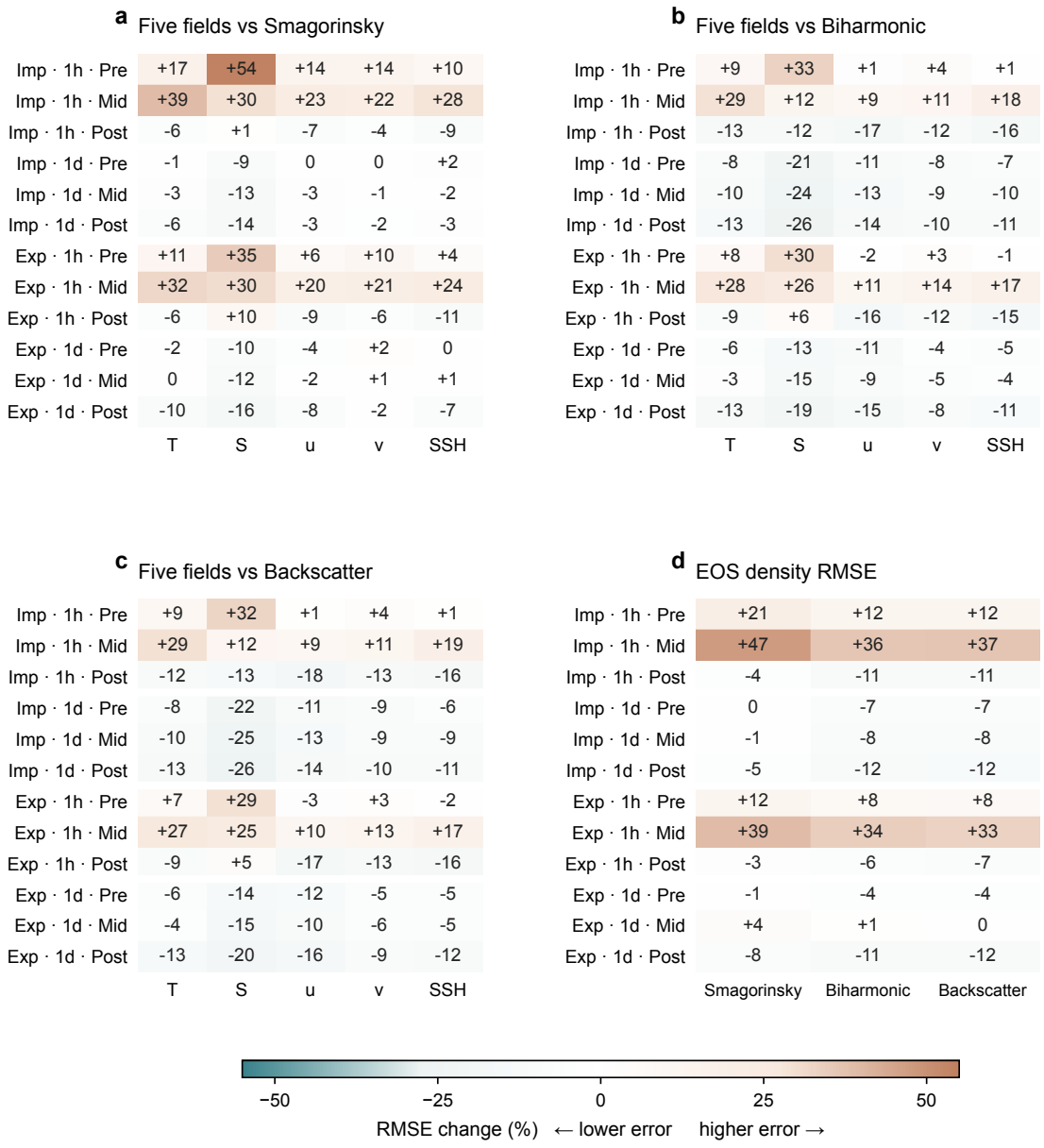


Figure S6: Forecast skill across all frozen stages, days 51–100. Percentage change in RMSE relative to each subgrid parameterization (`Smagorinsky`, `Biharmonic` and `Backscatter`) for all 12 DND configurations. Each score pools daily MSE over the 16 origins and the indicated window before taking its square root. Negative values denote lower error.

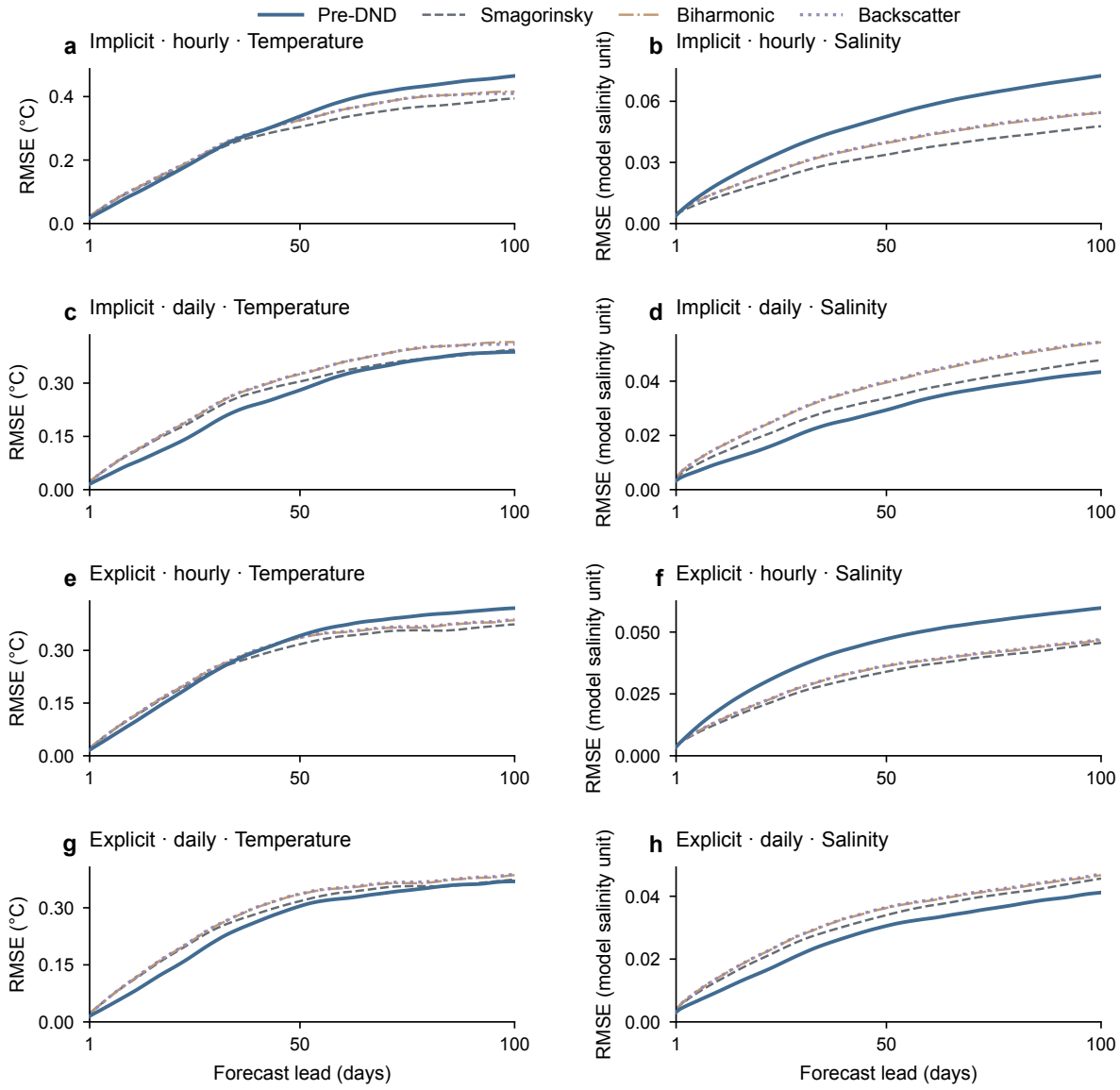


Figure S7: Pretrain-DND temperature and salinity error growth. Daily RMSE is the square root of mean MSE across 16 overlapping origins at each lead, over days 1–100. Curves show the two cores and correction intervals with matched physical parameterizations. They are unsmoothed; the origins are paired comparisons from one training seed.

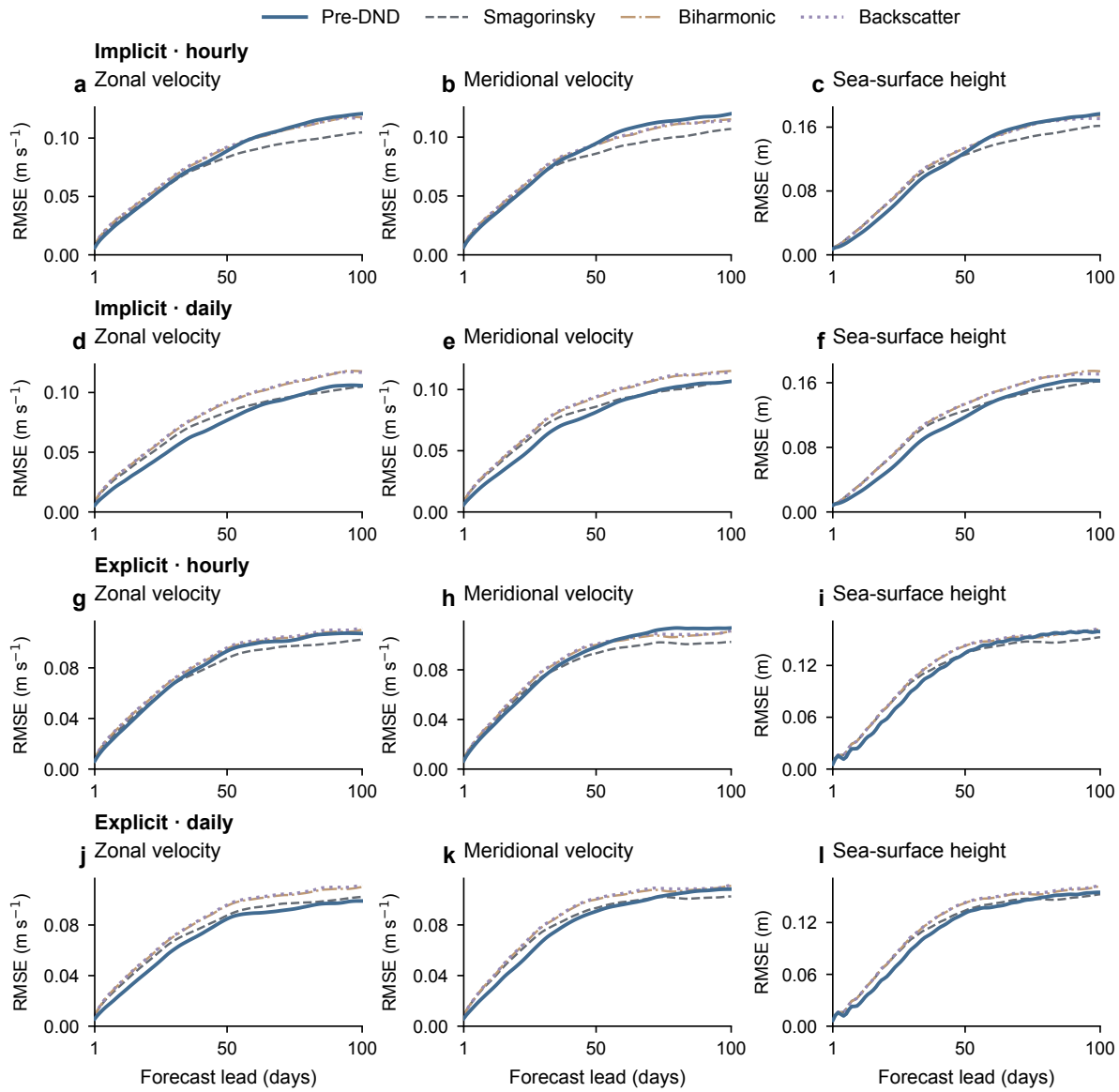


Figure S8: Pretrain-DND horizontal velocity and sea-surface height error growth. Daily RMSE is the square root of mean MSE across 16 overlapping origins at each lead, over days 1–100. Curves show the two cores and correction intervals with matched physical parameterizations. They are unsmoothed; the origins are paired comparisons from one training seed.

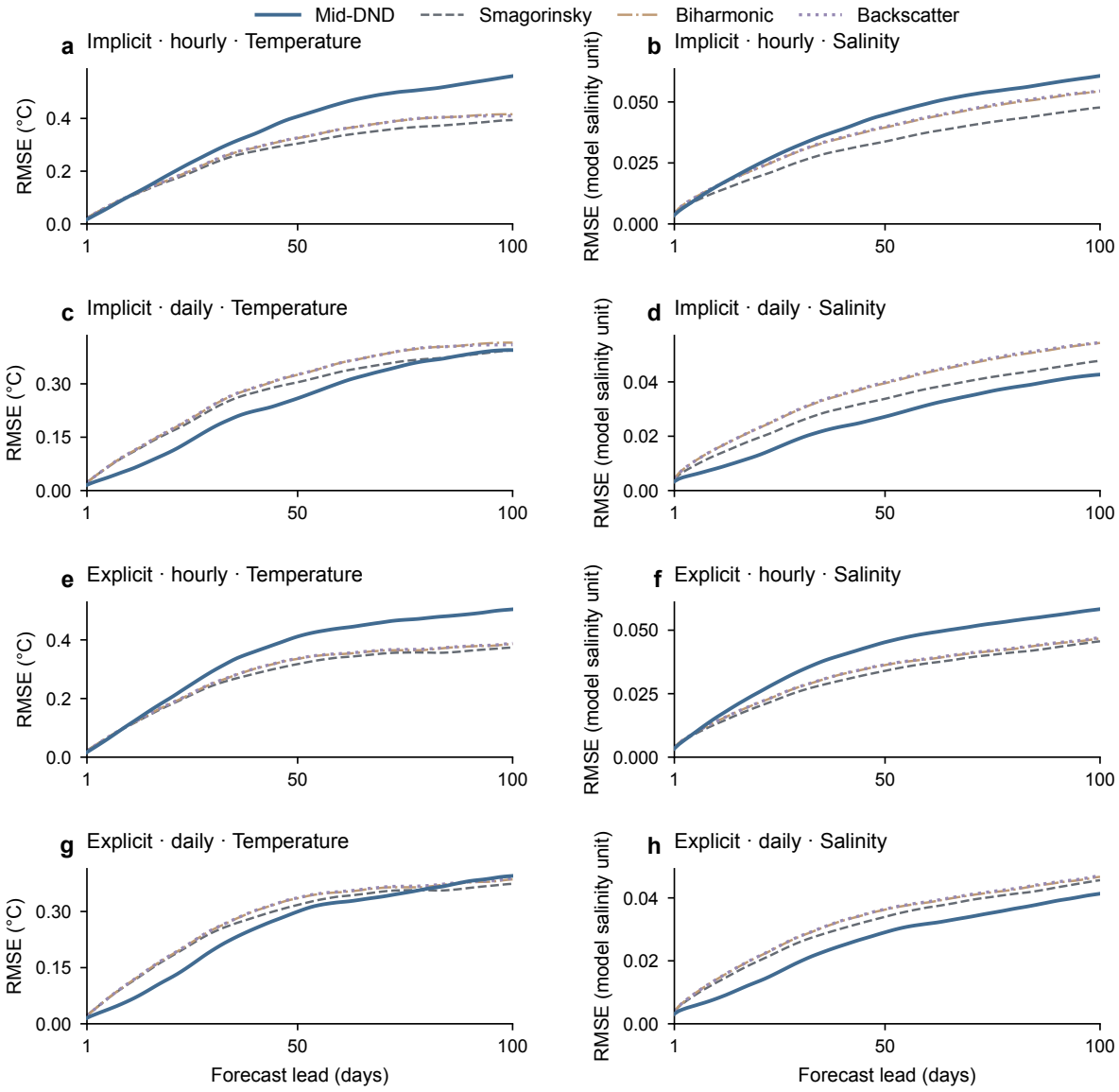


Figure S9: Midtrain-DND temperature and salinity error growth. Daily RMSE is the square root of mean MSE across 16 overlapping origins at each lead, over days 1–100. Curves show the two cores and correction intervals with matched physical parameterizations. They are unsmoothed; the origins are paired comparisons from one training seed.

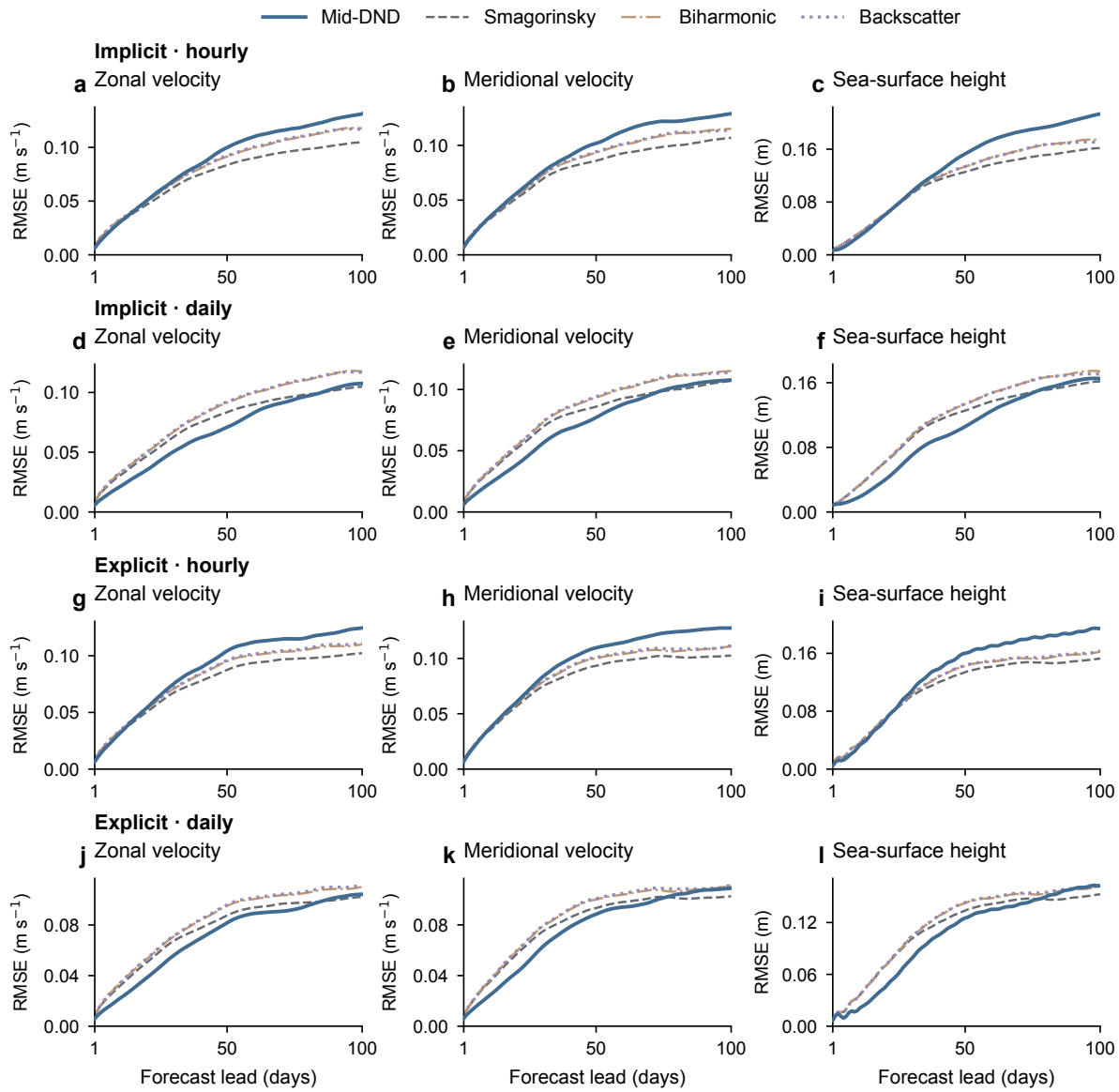


Figure S10: Midtrain-DND horizontal velocity and sea-surface height error growth. Daily RMSE is the square root of mean MSE across 16 overlapping origins at each lead, over days 1–100. Curves show the two cores and correction intervals with matched physical parameterizations. They are unsmoothed; the origins are paired comparisons from one training seed.

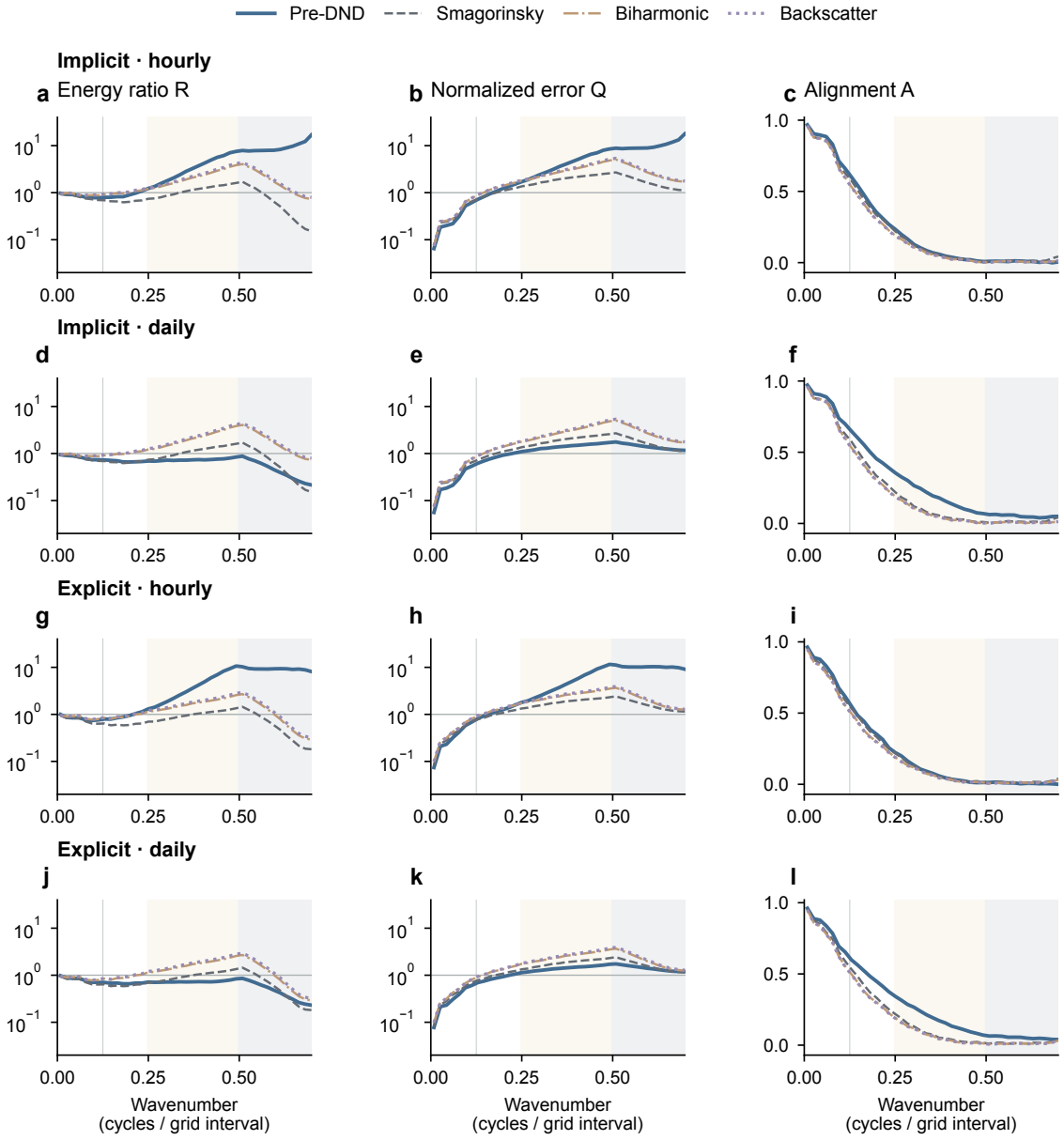


Figure S11: Pretrain-DND spectra, days 1–50. Prediction/reference energy ratio R, error/reference energy ratio Q, and normalized real cross-energy alignment A. The representation is sigma-thickness-weighted layerwise velocity energies. Energies are pooled over days and origins before ratios are formed. Wavenumber is cycles per grid interval; guides mark 0.125, 0.25 and 0.5. Near-grid and radial corner bins remain visible. Layerwise and depth-mean spectra measure different aspects of the circulation.

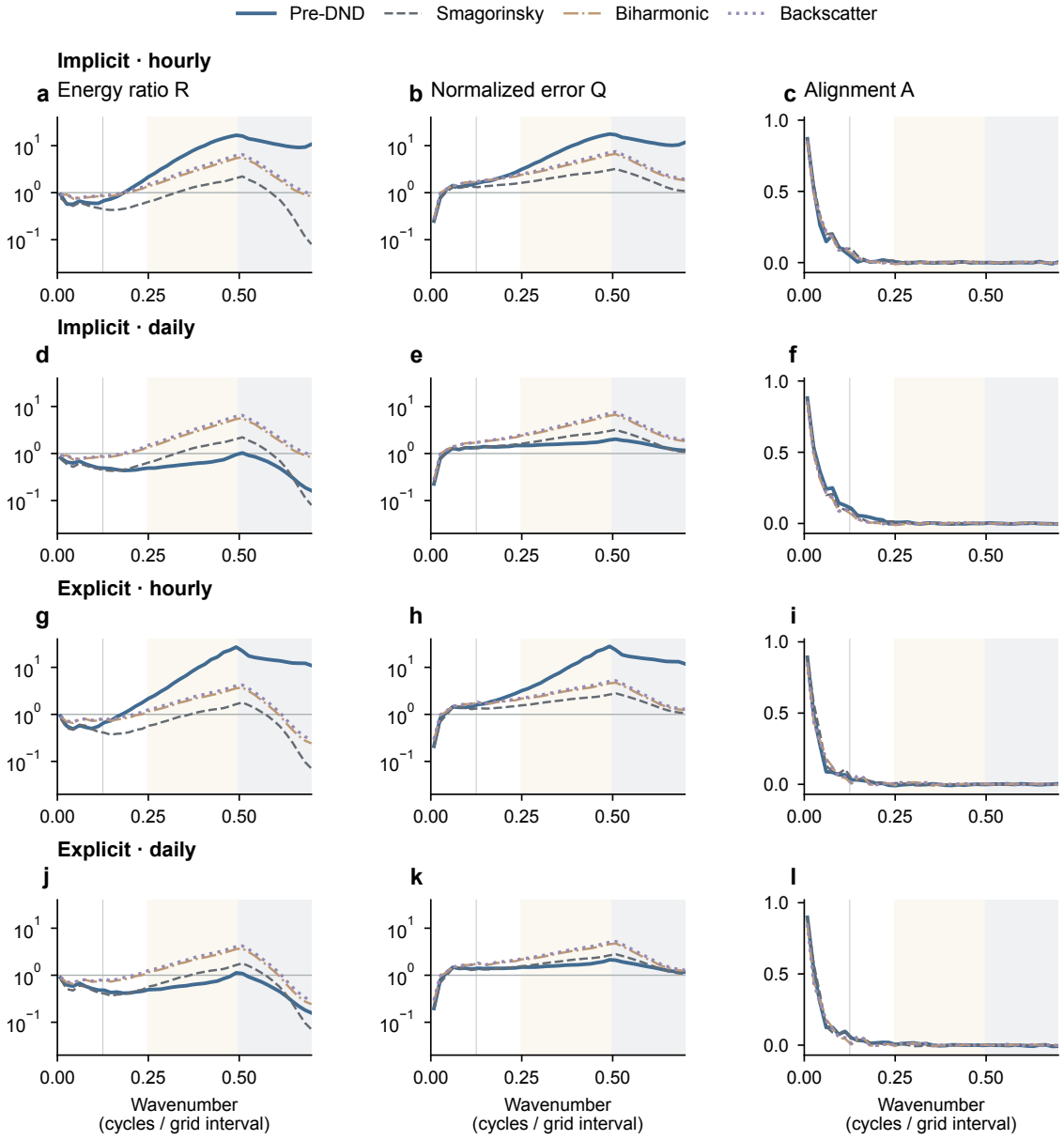


Figure S12: Pretrain-DND spectra, days 51–100. Prediction/reference energy ratio R, error/reference energy ratio Q, and normalized real cross-energy alignment A. The representation is sigma-thickness-weighted layerwise velocity energies. Energies are pooled over days and origins before ratios are formed. Wavenumber is cycles per grid interval; guides mark 0.125, 0.25 and 0.5. Near-grid and radial corner bins remain visible. Layerwise and depth-mean spectra measure different aspects of the circulation.

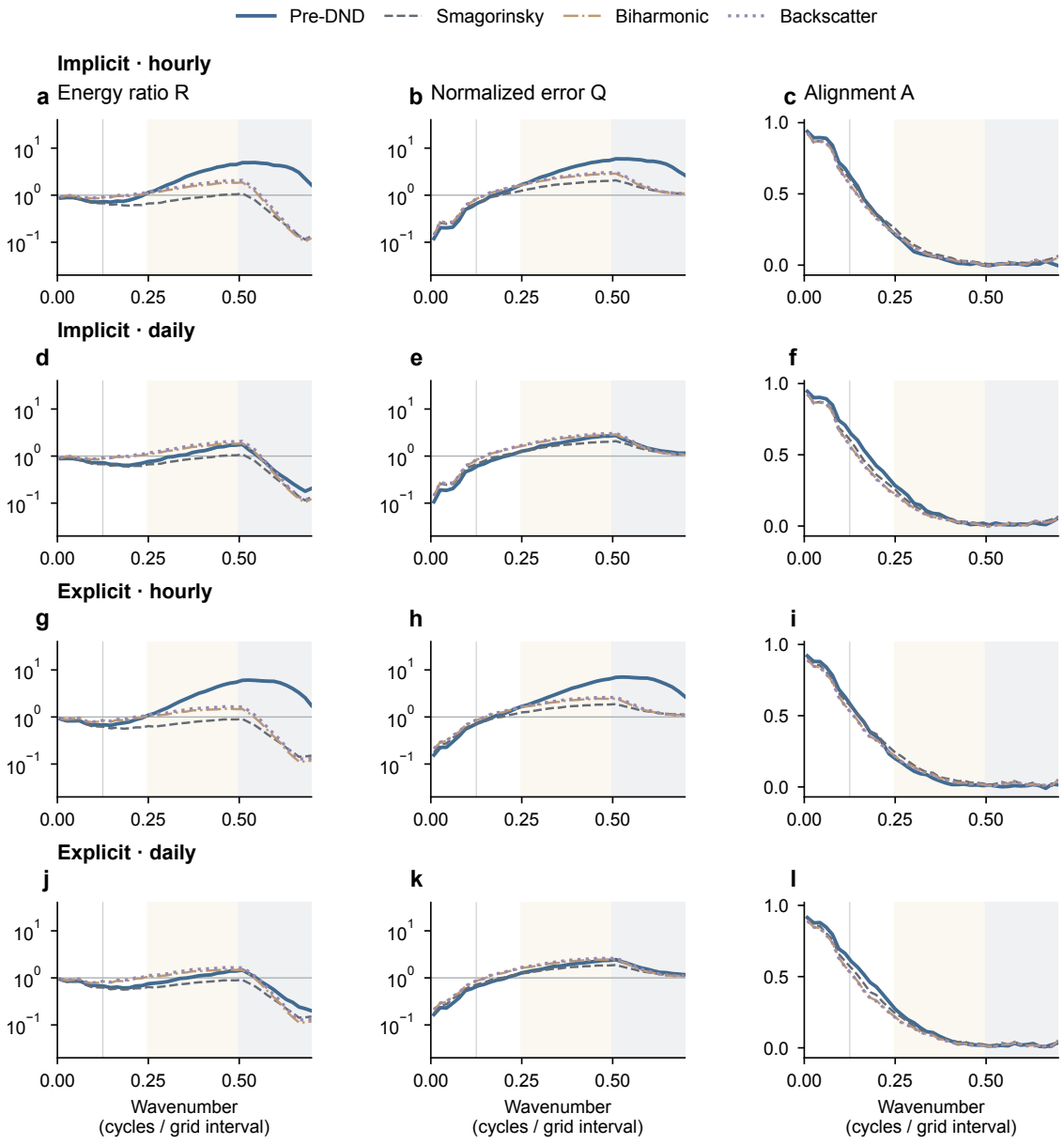


Figure S13: Pretrain-DND spectra, days 1–50. Prediction/reference energy ratio R, error/reference energy ratio Q, and normalized real cross-energy alignment A. The representation is sigma-weighted depth-mean velocity. Energies are pooled over days and origins before ratios are formed. Wavenumber is cycles per grid interval; guides mark 0.125, 0.25 and 0.5. Near-grid and radial corner bins remain visible. Layerwise and depth-mean spectra measure different aspects of the circulation.

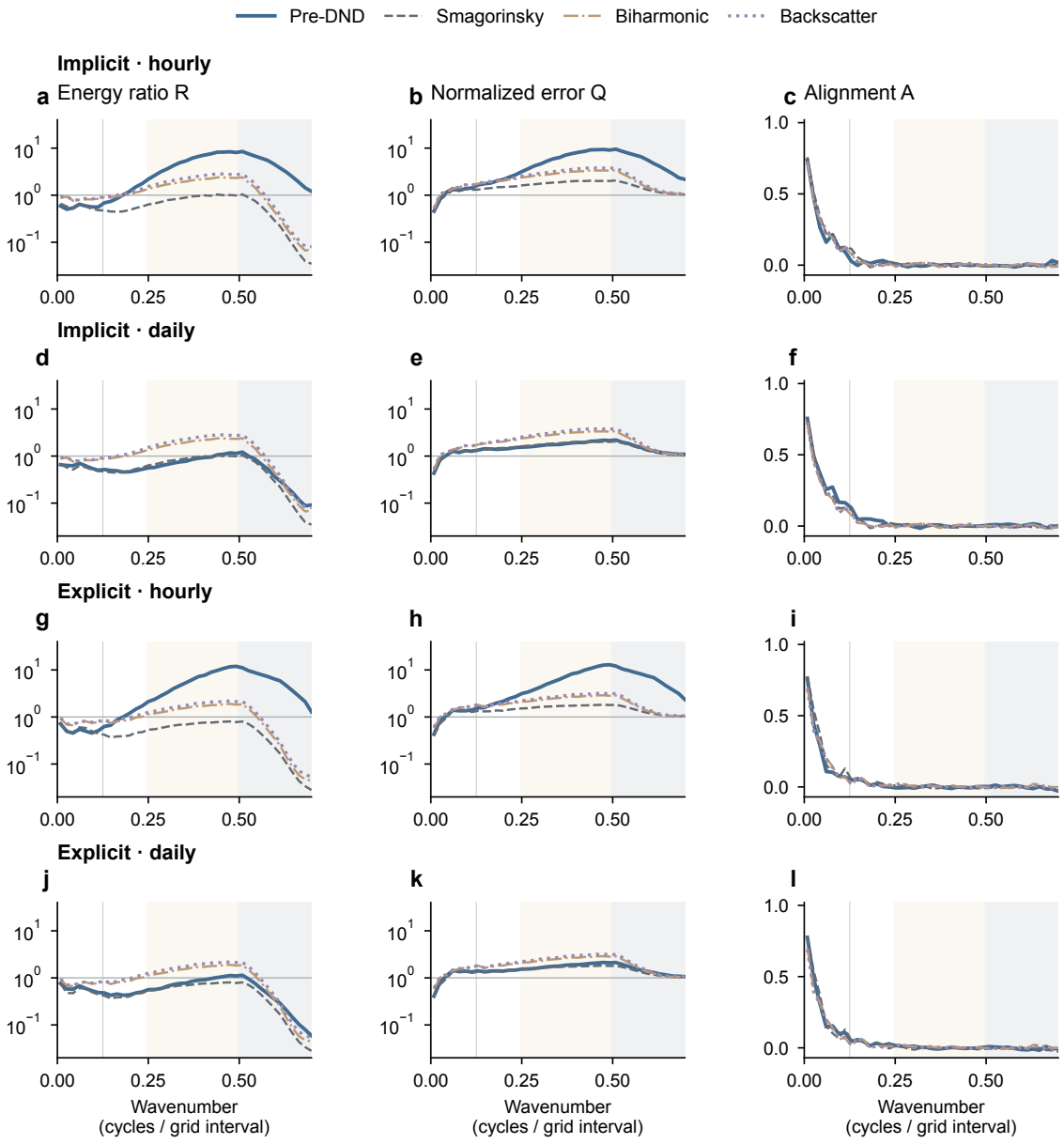


Figure S14: Pretrain-DND spectra, days 51–100. Prediction/reference energy ratio R, error/reference energy ratio Q, and normalized real cross-energy alignment A. The representation is sigma-weighted depth-mean velocity. Energies are pooled over days and origins before ratios are formed. Wavenumber is cycles per grid interval; guides mark 0.125, 0.25 and 0.5. Near-grid and radial corner bins remain visible. Layerwise and depth-mean spectra measure different aspects of the circulation.

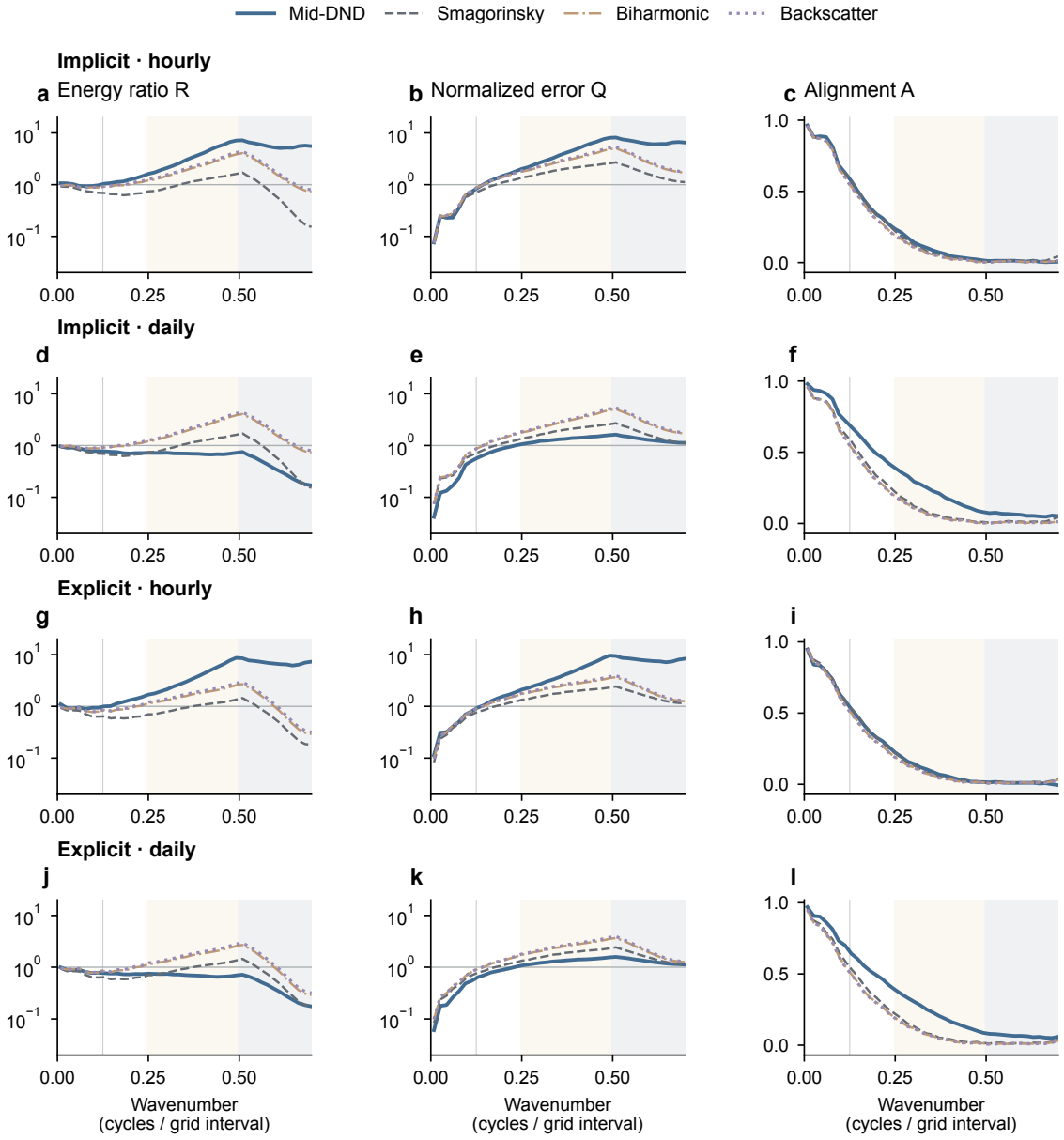


Figure S15: Midtrain-DND spectra, days 1–50. Prediction/reference energy ratio R, error/reference energy ratio Q, and normalized real cross-energy alignment A. The representation is sigma-thickness-weighted layerwise velocity energies. Energies are pooled over days and origins before ratios are formed. Wavenumber is cycles per grid interval; guides mark 0.125, 0.25 and 0.5. Near-grid and radial corner bins remain visible. Layerwise and depth-mean spectra measure different aspects of the circulation.

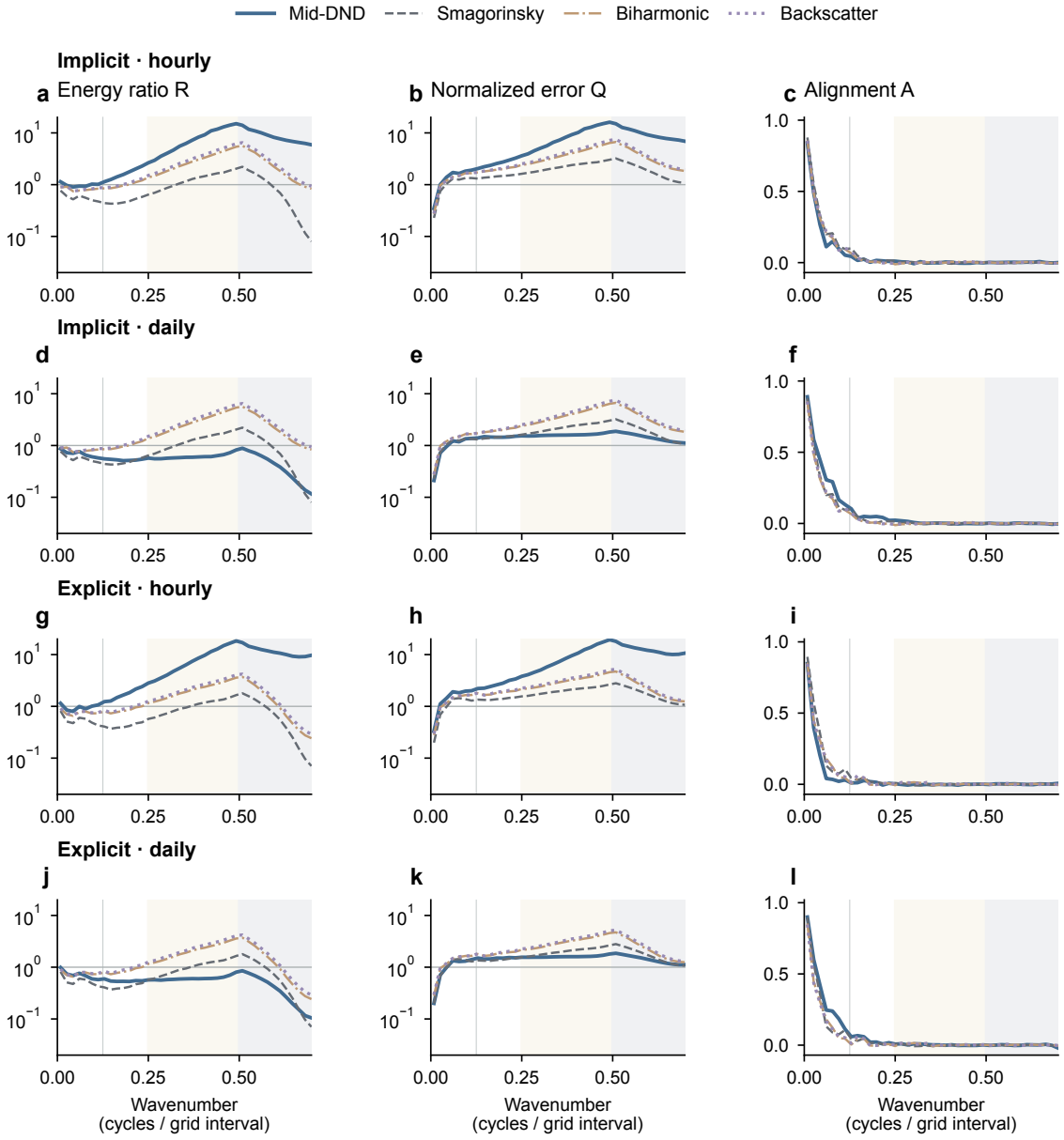


Figure S16: Midtrain-DND spectra, days 51–100. Prediction/reference energy ratio R, error/reference energy ratio Q, and normalized real cross-energy alignment A. The representation is sigma-thickness-weighted layerwise velocity energies. Energies are pooled over days and origins before ratios are formed. Wavenumber is cycles per grid interval; guides mark 0.125, 0.25 and 0.5. Near-grid and radial corner bins remain visible. Layerwise and depth-mean spectra measure different aspects of the circulation.

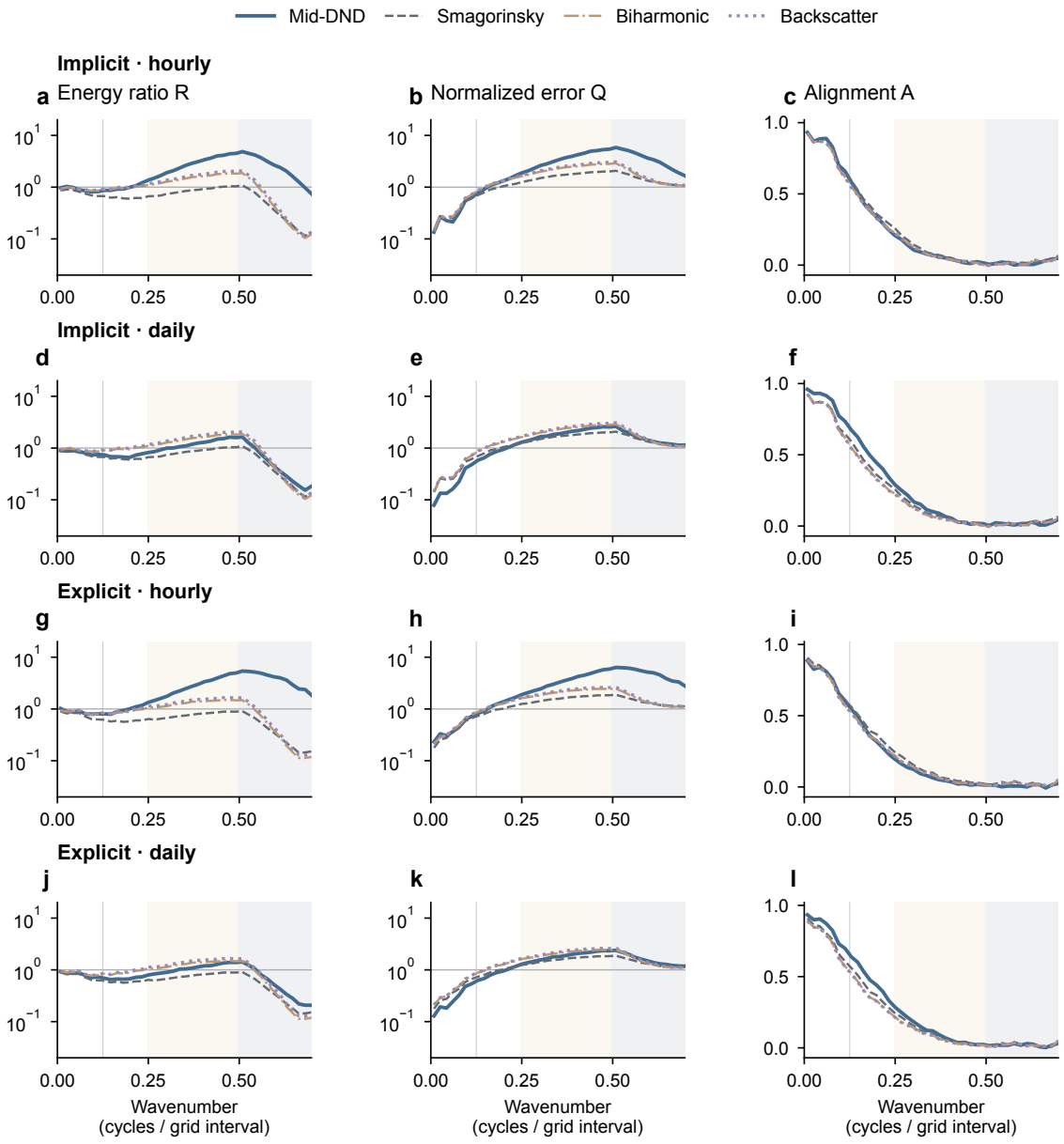


Figure S17: Midtrain-DND spectra, days 1–50. Prediction/reference energy ratio R, error/reference energy ratio Q, and normalized real cross-energy alignment A. The representation is sigma-weighted depth-mean velocity. Energies are pooled over days and origins before ratios are formed. Wavenumber is cycles per grid interval; guides mark 0.125, 0.25 and 0.5. Near-grid and radial corner bins remain visible. Layerwise and depth-mean spectra measure different aspects of the circulation.

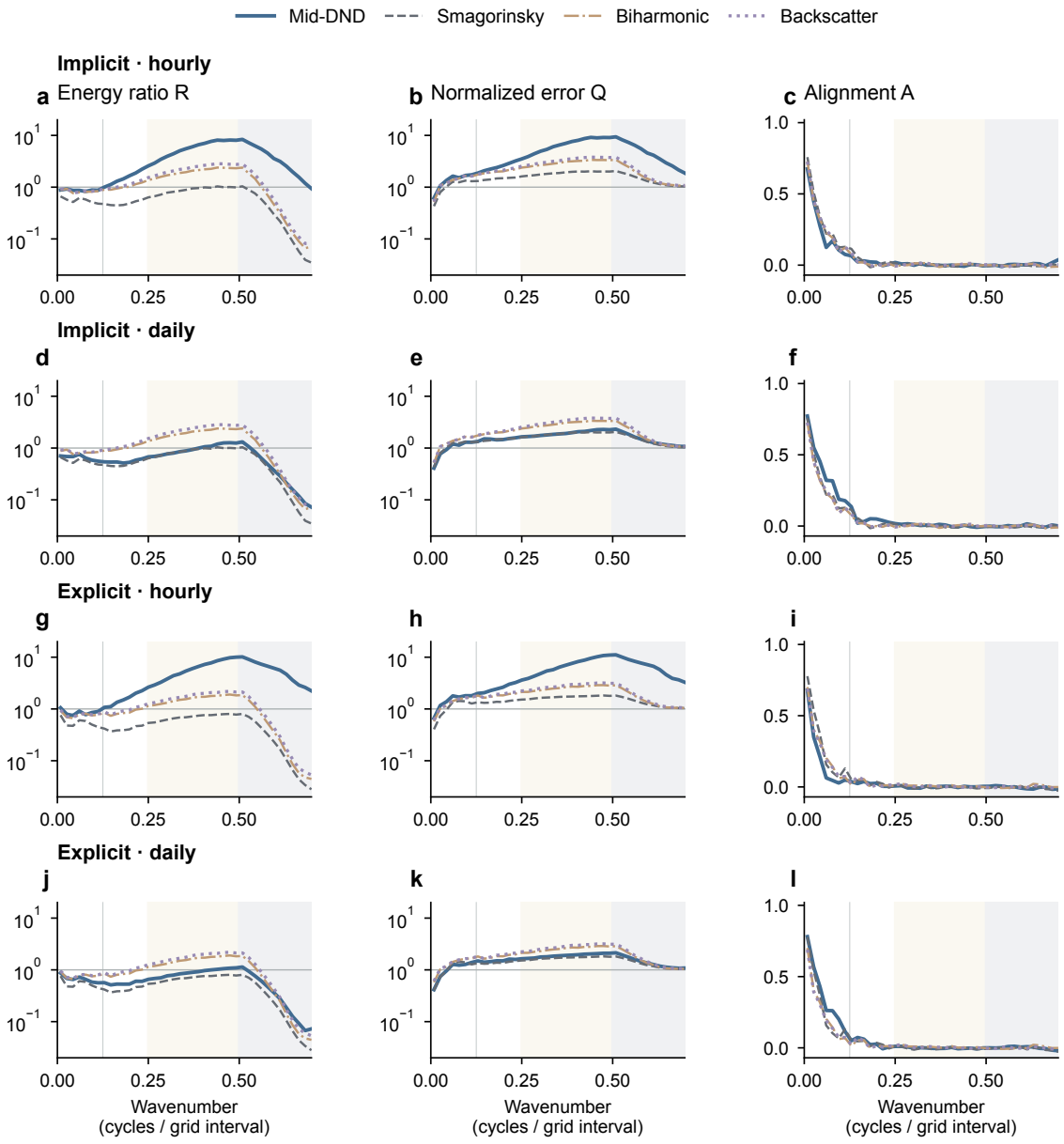


Figure S18: Midtrain-DND spectra, days 51–100. Prediction/reference energy ratio R, error/reference energy ratio Q, and normalized real cross-energy alignment A. The representation is sigma-weighted depth-mean velocity. Energies are pooled over days and origins before ratios are formed. Wavenumber is cycles per grid interval; guides mark 0.125, 0.25 and 0.5. Near-grid and radial corner bins remain visible. Layerwise and depth-mean spectra measure different aspects of the circulation.

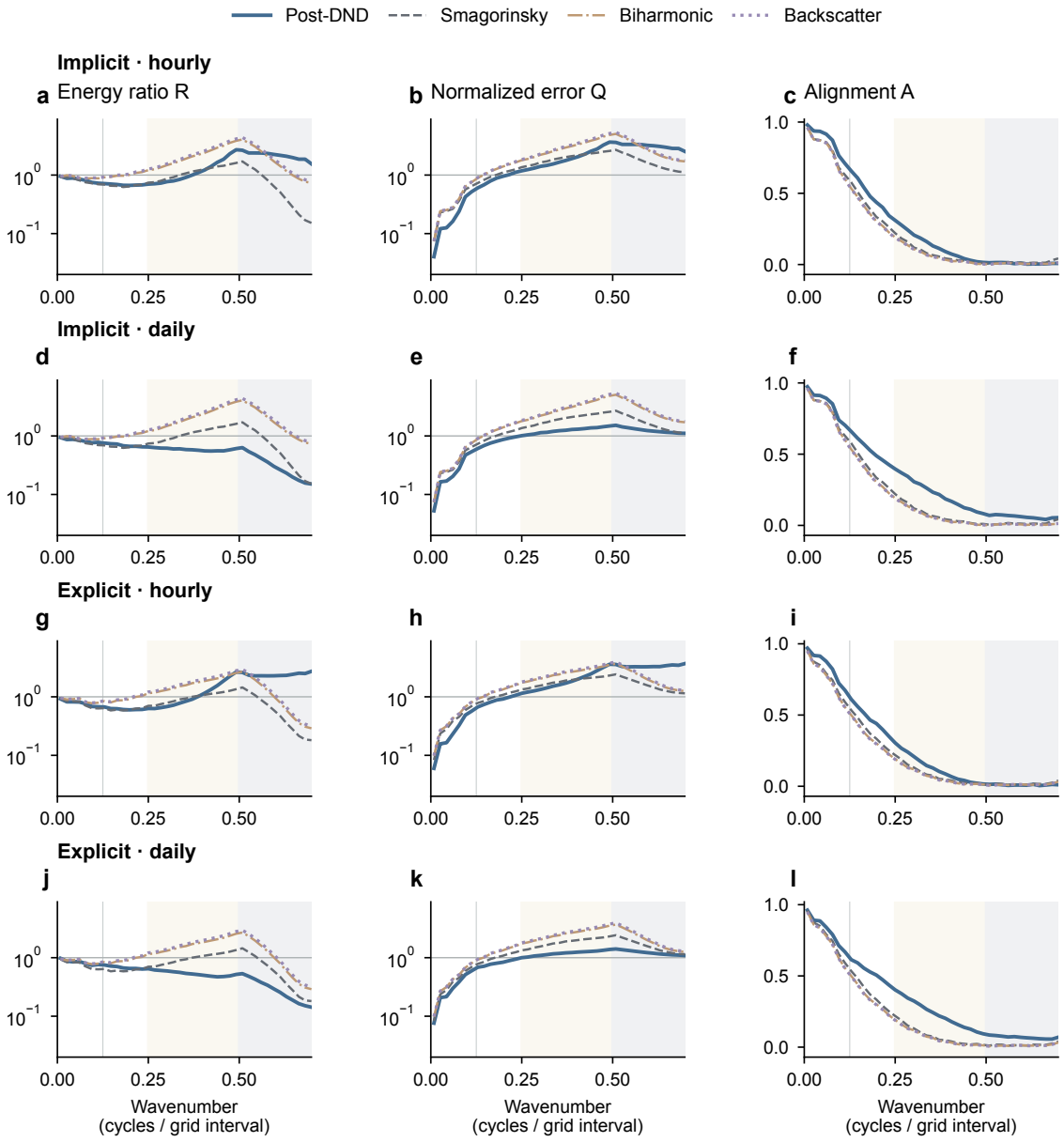


Figure S19: Posttrain-DND spectra, days 1–50. Prediction/reference energy ratio R, error/reference energy ratio Q, and normalized real cross-energy alignment A. The representation is sigma-thickness-weighted layerwise velocity energies. Energies are pooled over days and origins before ratios are formed. Wavenumber is cycles per grid interval; guides mark 0.125, 0.25 and 0.5. Near-grid and radial corner bins remain visible. Layerwise and depth-mean spectra measure different aspects of the circulation.

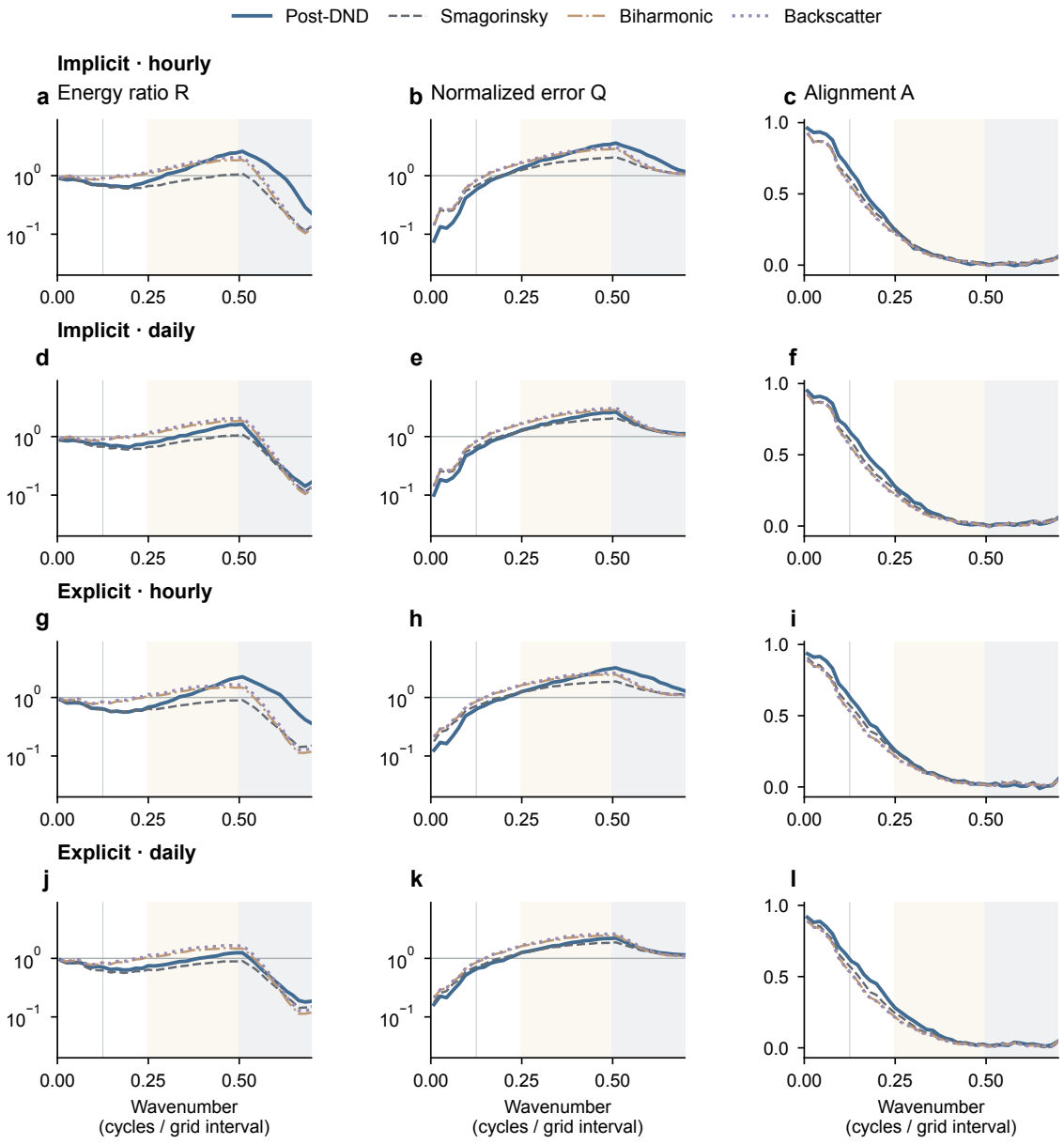


Figure S20: Posttrain-DND spectra, days 1–50. Prediction/reference energy ratio R, error/reference energy ratio Q, and normalized real cross-energy alignment A. The representation is sigma-weighted depth-mean velocity. Energies are pooled over days and origins before ratios are formed. Wavenumber is cycles per grid interval; guides mark 0.125, 0.25 and 0.5. Near-grid and radial corner bins remain visible. Layerwise and depth-mean spectra measure different aspects of the circulation.